1 **(a)** 13 **[B1]**

(b) Finds $f(2) = 1$ **[M1]**

-2 **[A1]**

(c) Sets $y = 5 - 2x$ and rearranges to $2x = 5 - y$ **[M1]**

$\dfrac{5-x}{2}$ **[A1]**

[Total Marks: 5]

2 **(a)** Sets $y = \dfrac{4}{2x-3}$ and multiplies by denominator to obtain $y(2x-3) = 4$ **[M1]**

Expands and rearranges to collect terms in x: $2xy = 4 + 3y$ **[M1]**

$\dfrac{4+3x}{2x}$ **[A1]**

(b) Finds $g(-3) = 2$ **[B1]**

Equates and attempts to solve: $\dfrac{4}{2x-3} = 2 \implies 4x - 6 = 4$ **[M1]**

2.5 (or $\dfrac{5}{2}$) **[A1]**

(c) Substitutes $f(x)$ into $g(x)$: $\dfrac{4}{2x-3} + 5$ **[M1]**

Expresses over a common denominator: $\dfrac{4 + 5(2x-3)}{2x-3}$ **[M1]**

$\dfrac{10x-11}{2x-3}$ **[A1]**

[Total Marks: 9]

3 **(a)** Sets $y = \dfrac{2x+5}{x-3}$ and rearranges to get $y(x-3) = 2x + 5$ **[M1]**

Expands and factorises to collect terms in x: $xy - 2x = 3y + 5 \implies x(y-2) = 3y + 5$ **[M1]**

$\dfrac{3x+5}{x-2}$ **[A1]**

(b) Substitutes $g(x)$ into $f(x)$: $\dfrac{2(3x-1)+5}{(3x-1)-3}$ **[M1]**

Expands and simplifies the numerator to $6x + 3$ **[M1]**

$\dfrac{6x+3}{3x-4}$ **[A1]**

(c) Sets up the equation $3\left(\dfrac{2x+5}{x-3}\right) - 1 = x$ **[M1]**

Eliminates the denominator to obtain $6x + 15 - (x-3) = x(x-3)$ (or equivalent) **[M1]**

Simplifies to form the correct 3-term quadratic equation: $x^2 - 8x - 18 = 0$ **[A1]**

Correct use of the quadratic formula $x = \dfrac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(-18)}}{2(1)}$ **[M1]**

9.83 and -1.83 **[A1]**

[Total Marks: 11]